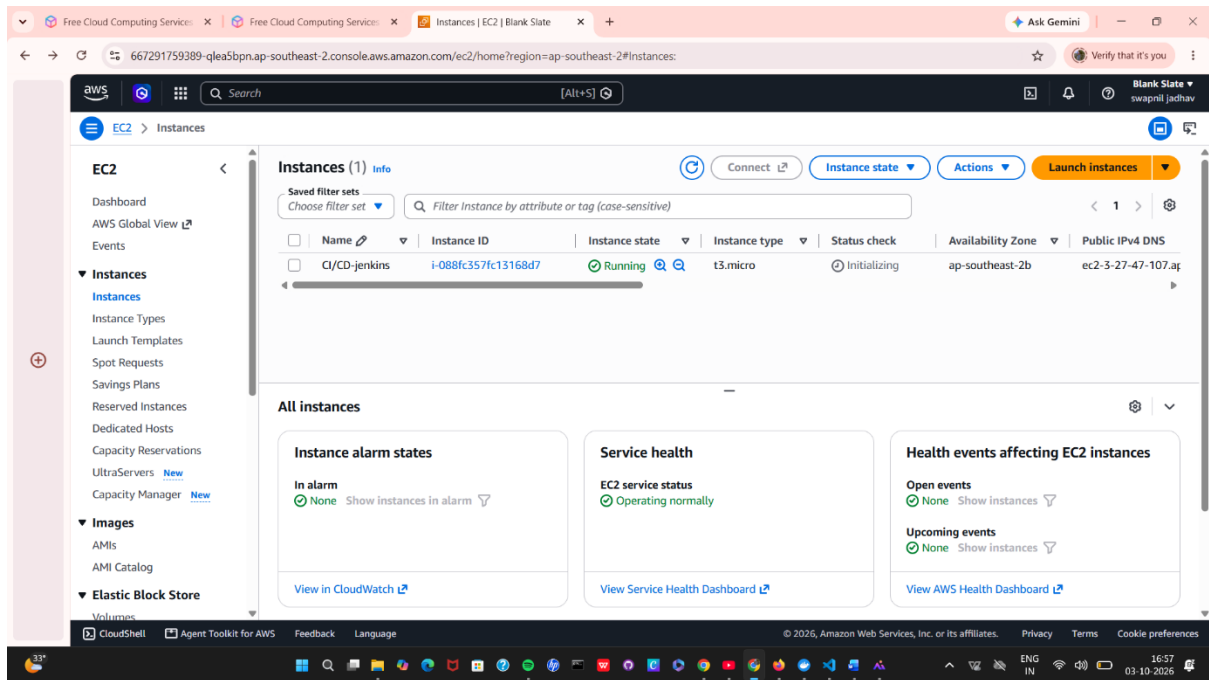


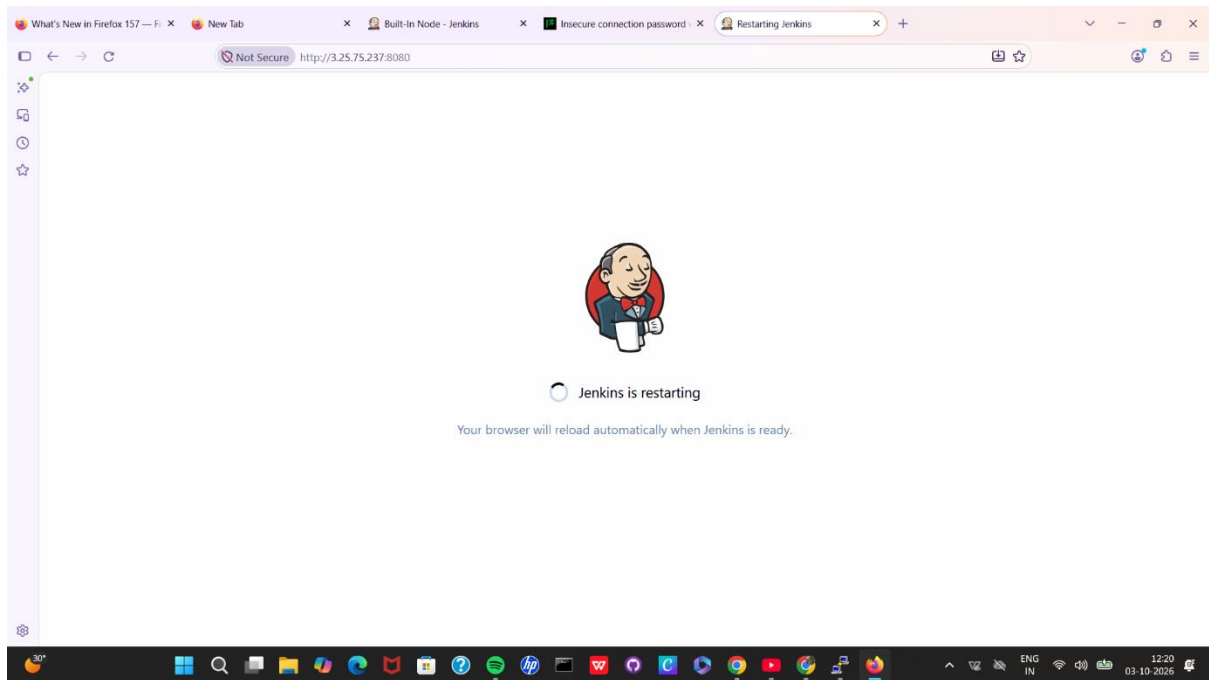
Create EC2 and login into it using SSH



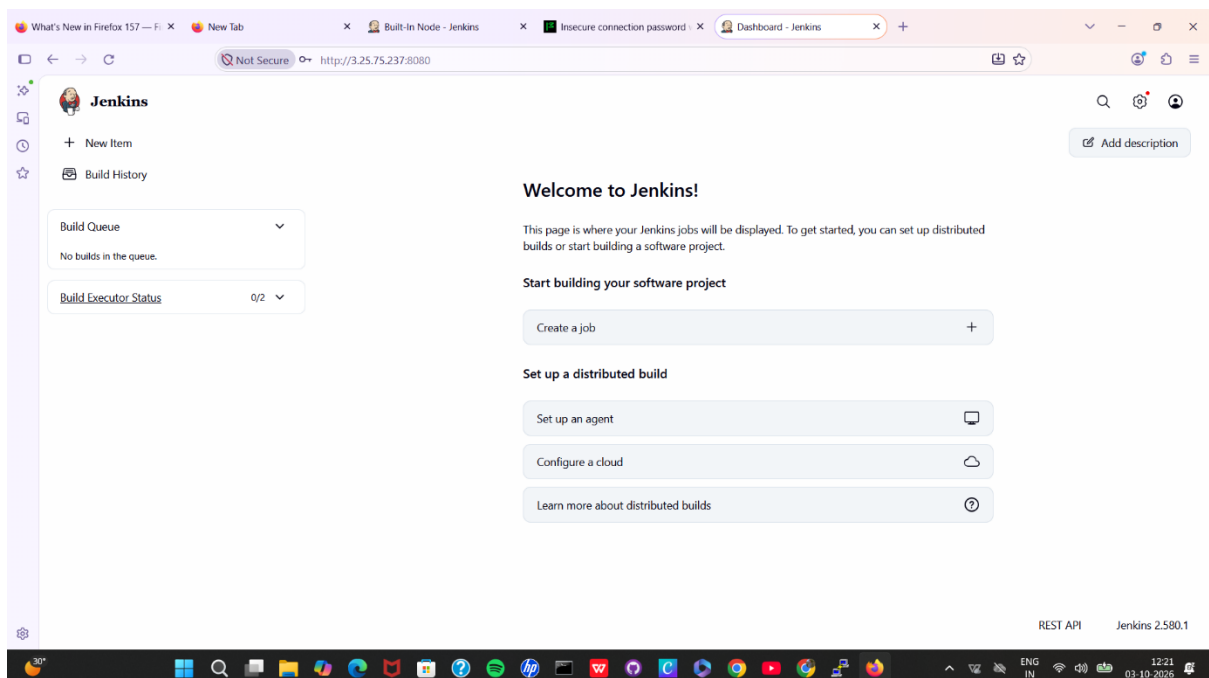
Install Jenkins in EC2

```
ubuntu@ip-172-31-42-176:~$ sudo apt update
Hit:1 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Get:4 https://pkg.jenkins.io/debian-stable binary/ InRelease
Get:5 https://pkg.jenkins.io/debian-stable binary/ Release [2044 B]
Get:6 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Get:7 https://pkg.jenkins.io/debian-stable binary/ Packages [31.8 kB]
Hit:8 http://security.ubuntu.com/ubuntu resolute-security InRelease
Fetched 34.7 kB in 9s (77.2 kB/s)
186 packages can be upgraded. Run 'apt list --upgradable' to see them.
Installing:
jenkins
Installing dependencies:
net-tools
Summary:
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 186
Download size: 53.9 MB
Space needed: 54.7 MB / 4282 MB available
Continue? [Y/n] y
Get:1 http://ap-southeast-2.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 net-tools [152 kB]
Get:2 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.580.1 [53.7 MB]
Fetched 53.9 MB in 9s (6127 kB/s)
Selecting previously unselected package net-tools.
(Reading database ... 37687 files and directories currently installed.)
Preparing to unpack .../net-tools-2.10-2ubuntu1_amd64v3.deb ...
Unpacking net-tools (2.10-2ubuntu1) ...
Selecting previously unselected package jenkins.
Preparing to unpack .../jenkins-2.580.1_all.deb ...
```

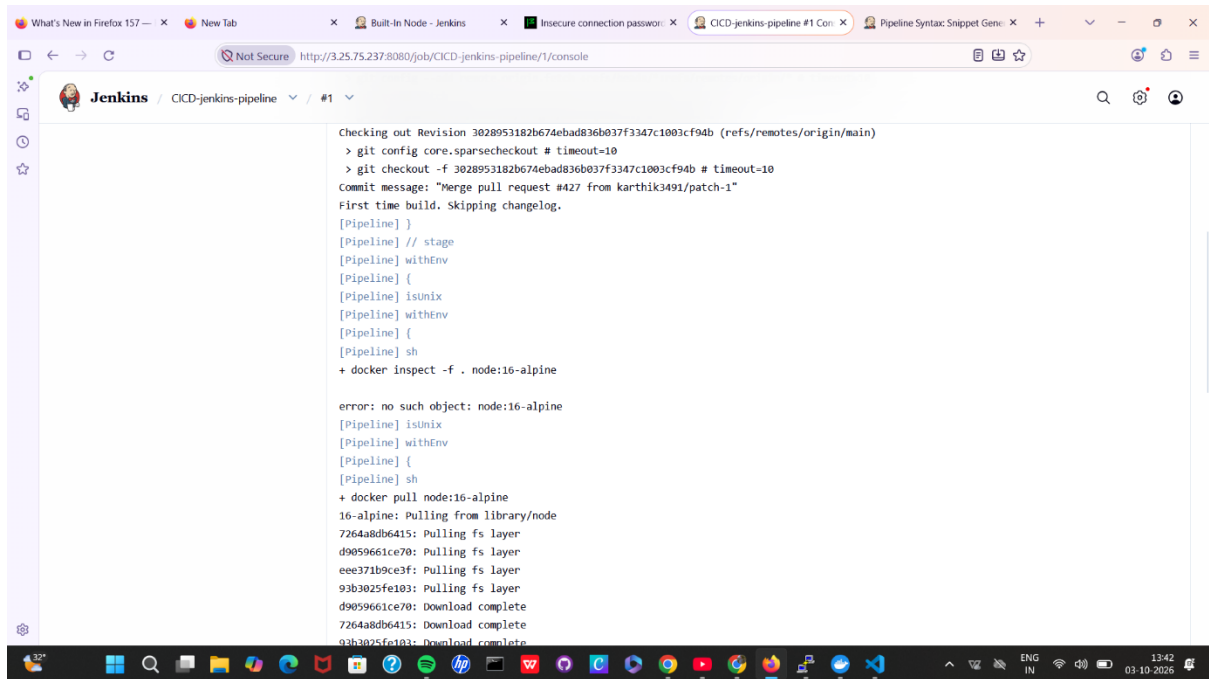
Login into Jenkins using Ec2 public IP and port number



Add plugins (Docker)



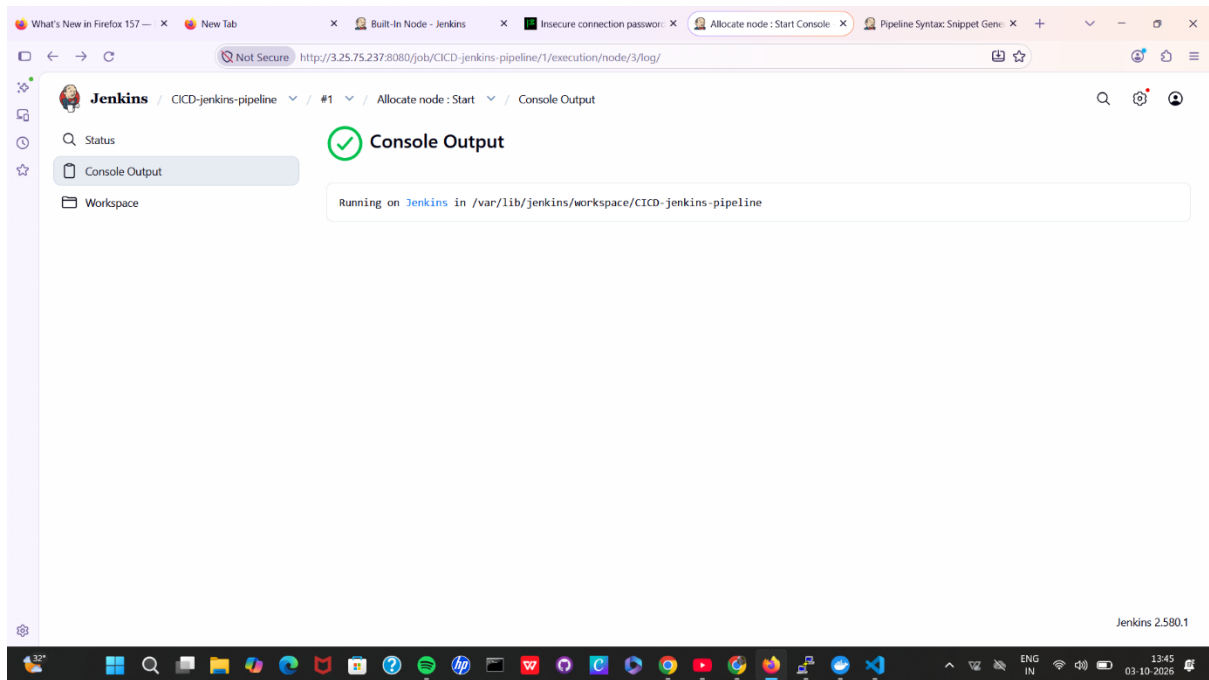
Successfully created the pipeline



The screenshot shows the Jenkins console output for a pipeline run. The output is as follows:

```
Checking out Revision 3028953182b674ebad836b037f3347c1003cf94b (refs/remotes/origin/main)
> git config core.sparsecheckout # timeout=10
> git checkout -f 3028953182b674ebad836b037f3347c1003cf94b # timeout=10
Commit message: "Merge pull request #427 from karthik3491/patch-1"
First time build. Skipping changelog.
[Pipeline] }
[Pipeline] // stage
[Pipeline] withEnv
[Pipeline] {
[Pipeline] isUnix
[Pipeline] withEnv
[Pipeline] {
[Pipeline] sh
+ docker inspect -f . node:16-alpine

error: no such object: node:16-alpine
[Pipeline] isUnix
[Pipeline] withEnv
[Pipeline] {
[Pipeline] sh
+ docker pull node:16-alpine
16-alpine: Pulling from library/node
7264a8db6415: Pulling fs layer
d9059661ce70: Pulling fs layer
eee371b9ce3f: Pulling fs layer
93b3025fe103: Pulling fs layer
d9059661ce70: Download complete
7264a8db6415: Download complete
93b3025fe103: Download complete
```



The screenshot shows the Jenkins console output for a pipeline run. The output is as follows:

```
Running on Jenkins in /var/lib/jenkins/workspace/CICD-jenkins-pipeline
```

The screenshot also shows the Jenkins interface with the "Console Output" tab selected. The output is displayed in a scrollable area. The Jenkins version is 2.580.1.

Used docker as agent in Jenkins project setup and I found tis as very useful in terms of cost and also in terms of efficiency.

```
ubuntu@ip-172-31-42-176:~$  
ubuntu@ip-172-31-42-176:~$  
ubuntu@ip-172-31-42-176:~$ sudo docker ps  
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES  
ubuntu@ip-172-31-42-176:~$
```

